

ECE 222 Homework #1

Due Thursday, September 10, 2026 at 11:59 pm on Canvas

Scan or photograph and save as a single .pdf file.

Name your file “HW_No_1_Last Name.pdf”

Do problems 3.36, 3.37, 3.42, 3.44, 3.58

3.36 Find the gradient of the following scalar functions:

- a. $T = 3/(x^2 + z^2)$
- b. $V = xy^2z^4$
- c. $U = z \cos \phi / (1 + r^2)$
- d. $W = e^{-R} \sin \theta$
- *e. $S = 4x^2e^{-z} + y^3$
- f. $N = r^2 \cos^2 \phi$
- g. $M = R \cos \theta \sin \phi$

3.37 For each of the following scalar fields, obtain an analytical solution for ∇T and generate a corresponding arrow representation.

- a. $T = 10 + x$, for $-10 \leq x \leq 10$
- *b. $T = x^2$, for $-10 \leq x \leq 10$
- b. $T = 100 + xy$, for $-10 \leq x \leq 10$
- c. $T = x^2y^2$, for $-10 \leq x, y \leq 10$
- e. $T = 20 + x + y$, for $-10 \leq x, y \leq 10$
- f. $T = 1 + \sin(\pi x/3)$, for $-10 \leq x \leq 10$
- *g. $T = 1 + \cos(\pi x/3)$, for $-10 \leq x \leq 10$
- h. $T = 15 + r \cos \phi$, for $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi. \end{cases}$
- i. $T = 15 + r \cos^2 \phi$, for $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi. \end{cases}$

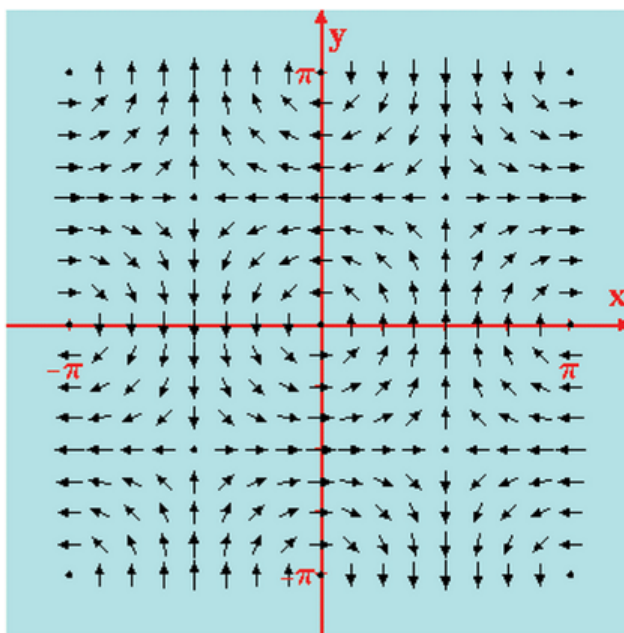
NOTE: See the figures in problem 3.44 for examples of “arrow representations”. You can make approximate sketches by hand.

3.42 For the scalar function $T = \frac{1}{2} e^{-r/5} \cos \phi$, determine its directional derivative along the radial direction $\hat{\mathbf{r}}$ and then evaluate it at $P = (2, \pi/4, 3)$.

3.44 Each of the following vector fields is displayed in Fig. P3.44 in the form of a vector representation. Determine $\nabla \cdot \mathbf{A}$ analytically and then compare the result with your expectations on the basis of the displayed pattern.

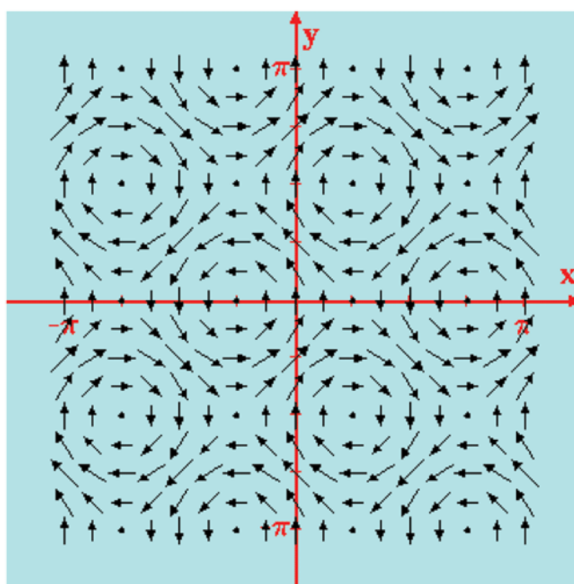
a. $\mathbf{A} = -\hat{x} \cos x \sin y + \hat{y} \sin x \cos y$, for $-\pi \leq x, y \leq \pi$

Figure P3.44(a)



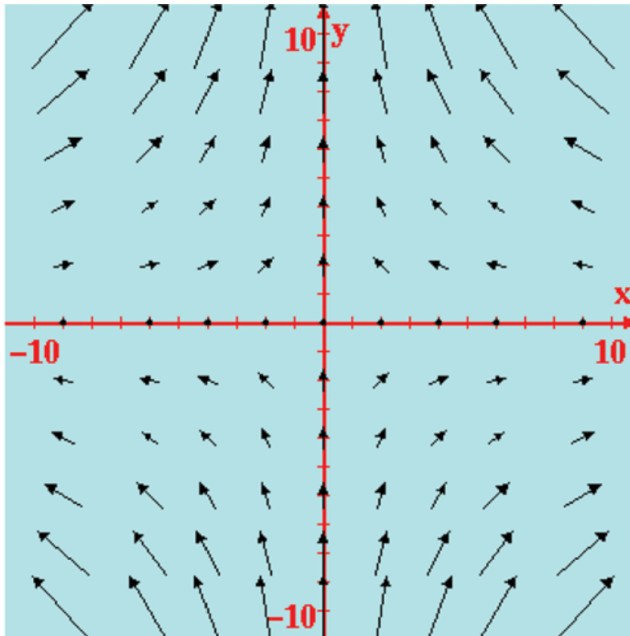
b. $\mathbf{A} = -\hat{x} \sin 2y + \hat{y} \cos 2x$, for $-\pi \leq x, y \leq \pi$

Figure P3.44(b)



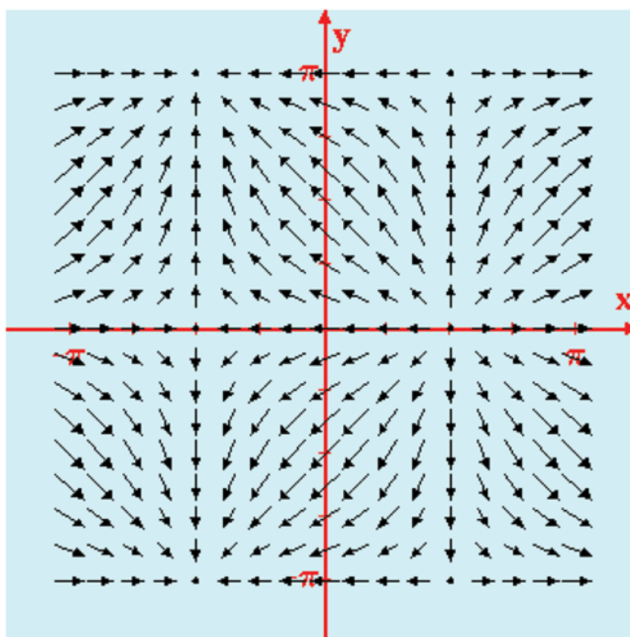
c. $\mathbf{A} = -\hat{\mathbf{x}} xy + \hat{\mathbf{y}} y^2$, for $-10 \leq x, y \leq 10$

Figure P3.44(c)



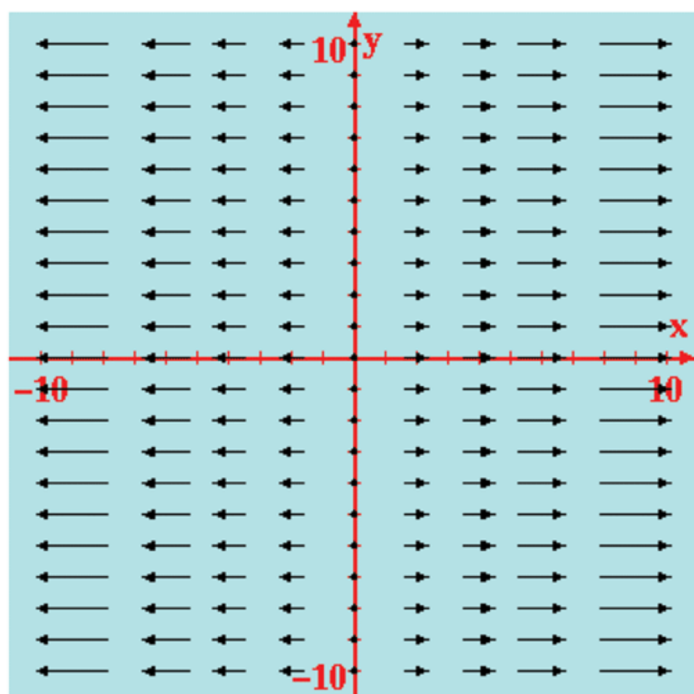
d. $\mathbf{A} = -\hat{\mathbf{x}} \cos x + \hat{\mathbf{y}} \sin y$, for $-\pi \leq x, y \leq \pi$

Figure P3.44(d)



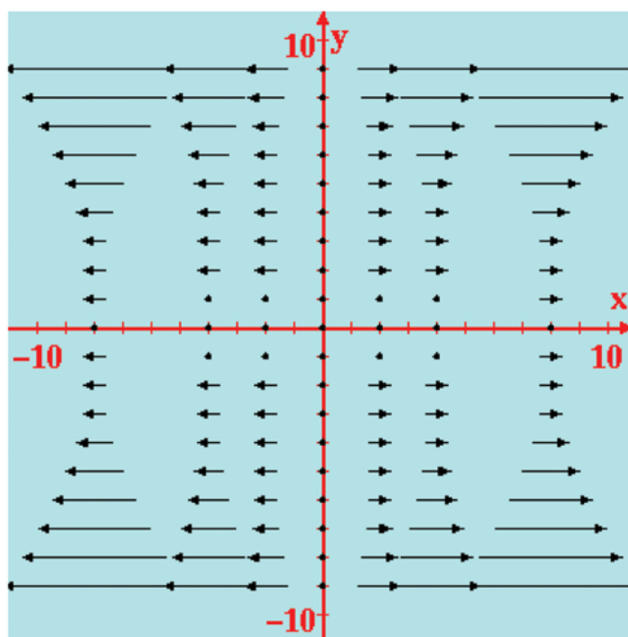
e. $\mathbf{A} = \hat{\mathbf{x}} x$, for $-10 \leq x \leq 10$

Figure P3.44(e)



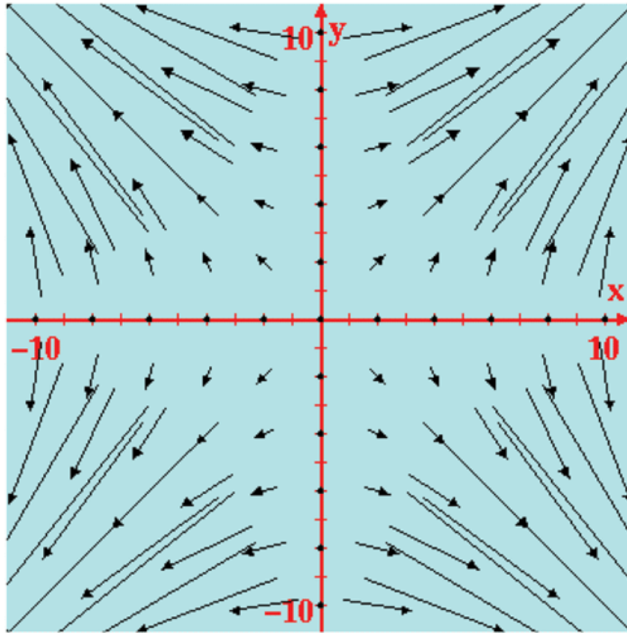
f. $\mathbf{A} = \hat{\mathbf{x}} xy^2$, for $-10 \leq x, y \leq 10$

Figure P3.44(f)



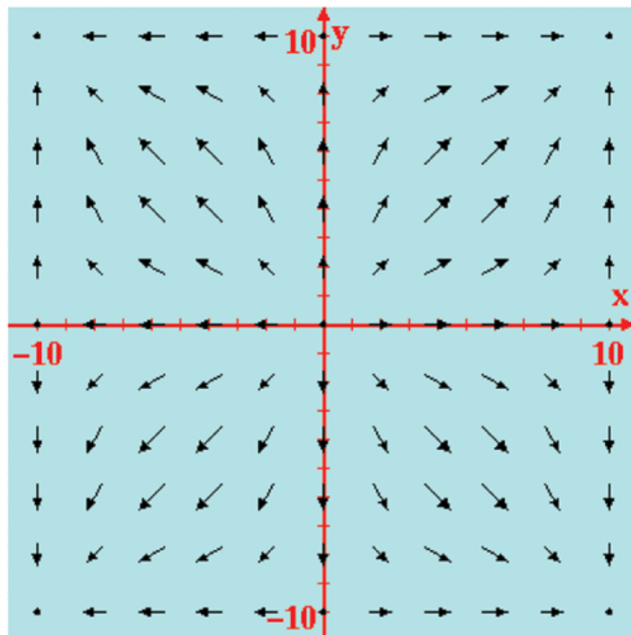
g. $\mathbf{A} = \hat{\mathbf{x}} xy^2 + \hat{\mathbf{y}} x^2y$, for $-10 \leq x, y \leq 10$

Figure P3.44(g)



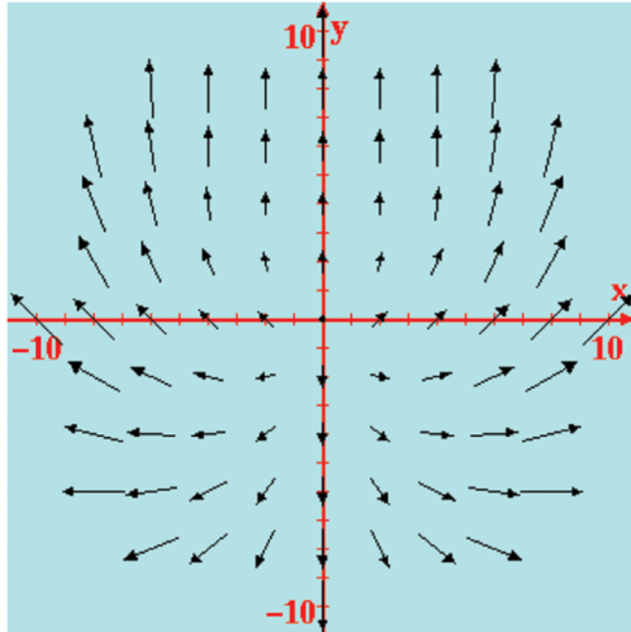
h. $\mathbf{A} = \hat{\mathbf{x}} \sin\left(\frac{\pi x}{10}\right) + \hat{\mathbf{y}} \sin\left(\frac{\pi y}{10}\right)$, for $-10 \leq x, y \leq 10$

Figure P3.44(h)



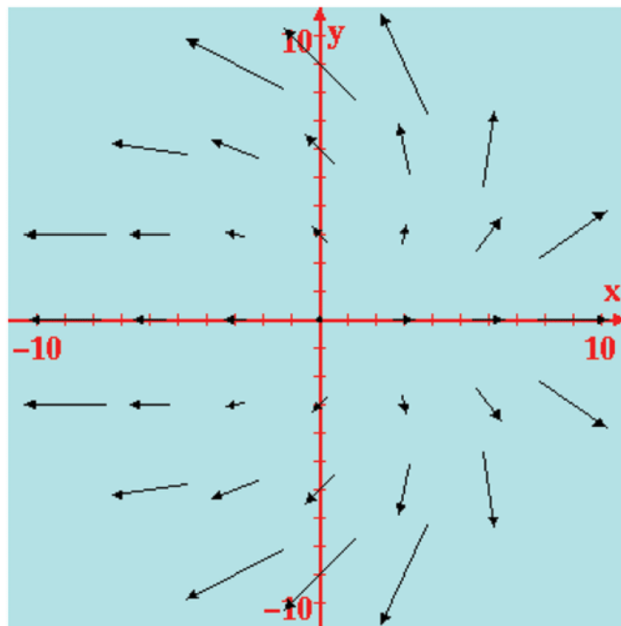
i. $\mathbf{A} = \hat{\mathbf{r}} r + \hat{\boldsymbol{\phi}} r \cos \phi$, for $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi. \end{cases}$

Figure P3.44(i)



j. $\mathbf{A} = \hat{\mathbf{r}} r^2 + \hat{\boldsymbol{\phi}} r^2 \sin \phi$, for $\begin{cases} 0 \leq r \leq 10 \\ 0 \leq \phi \leq 2\pi. \end{cases}$

Figure P3.44(j)



3.58 Determine if each of the following vector fields is solenoidal, conservative, or both:

- * a. $\mathbf{A} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}2xy$
- b. $\mathbf{B} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}y^2 + \hat{\mathbf{z}}2z$
- c. $\mathbf{C} = \hat{\mathbf{r}}(\sin \phi)/r^2 + \hat{\phi}(\cos \phi)/r^2$
- * d. $\mathbf{D} = \hat{\mathbf{R}}/R$
- e. $\mathbf{E} = \hat{\mathbf{r}} \left(3 - \frac{r}{1+r} \right) + \hat{\mathbf{z}}z$
- f. $\mathbf{F} = (\hat{\mathbf{x}}y + \hat{\mathbf{y}}x)/(x^2 + y^2)$
- g. $\mathbf{G} = \hat{\mathbf{x}}(x^2 + z^2) - \hat{\mathbf{y}}(y^2 + x^2) - \hat{\mathbf{z}}(y^2 + z^2)$
- * h. $\mathbf{H} = \hat{\mathbf{R}}(Re^{-R})$

Answers to starred problems:

3.36 $\square$ (e) $\nabla S = \hat{\mathbf{x}}8xe^{-z} + \hat{\mathbf{y}}3y^2 - \hat{\mathbf{z}}4x^2e^{-z}$

3.37 $\square$ (b) $\nabla T = \hat{\mathbf{x}}2x$

(g) $\nabla T = -\hat{\mathbf{x}} \frac{2\pi}{6} \sin\left(\frac{\pi x}{3}\right)$

3.58 $\square$ (a) $\mathbf{A}$ is solenoidal, but not conservative.

(d) $\mathbf{D}$ is conservative, but not solenoidal.

(h) $\mathbf{H}$ is conservative, but not solenoidal.